The standard Euclidean algorithm finds the GCD by repeatedly dividing two numbers and using the remainder. The Extended Euclidean algorithm does the same thing, but it also finds two values, x and y, that satisfy the equation $ax + by = GCD(a, b)$. Both algorithms have a time complexity of $O(\log n)$, but the Extended Euclidean algorithm is usually a bit slower because it needs extra calculations. If we only need to find the GCD, calculate the least common multiple , simplify fractions, or check whether two numbers have a GCD of 1, the standard Euclidean algorithm is a better choice because it is simpler and faster. However, the Extended Euclidean algorithm is useful when we need to solve more complex math problems, such as finding modular inverses or working with encryption